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## Introduction to AI Ethics

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### Summary

This handout identifies [ethical risks](#) of [artificial intelligence](#) (AI), particularly generative large language models, in an academic context. Students are encouraged to investigate the ethical concerns relevant to their use of AI. [AI ethics](#) describe the use and development of AI in line with moral values. AI ethics are shaped by [Big Tech](#) companies, such as Apple, Google, and Microsoft, and other powerful actors [to influence how the public views AI](#) and [gain power](#). Through understanding AI ethics, students can learn about AI harm reduction. Harm reduction can help to reduce the [negative consequences or harms associated with risky behaviors](#), including the usage of AI, through education and making choices that lend to less harm.

This handout empowers students with knowledge about AI. When considering AI ethics practically, there are [three realms which should be considered](#). How does the use of AI affect the self? How does it affect others (such as those on whose work it was trained)? And how does it affect society (in terms of environmental impacts or the perpetuation of bias, for instance)?

This handout offers a framework for answering these questions, and in answering them, for reducing the harm of AI usage.

### TL;DR: Reducing Harm

Generative AI is harmful. If you choose to use AI models like ChatGPT, you can reduce [global](#), [learning](#), and [personal](#) harms by planning the number of prompts, scope, and time limits of AI use, and taking breaks from AI-generated content. To minimize [inequities](#) in AI outputs, script prompts to include traditionally-marginalized voices. To reduce [learning harms](#), review AI outputs thoroughly for errors and/or missing information, disclose use of AI and harm reduction measures, and avoid using AI-generated media to get around potential plagiarism violations.

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### TL;DR: Effective Uses of AI

Still, there are cases in which AI may be useful. The responsible utilization of AI requires identifying these cases, and avoiding over-reliance on AI tools for the reasons outlined above. One [study](#) shows that AI can improve productivity when it is applied to routine accounting tasks, for example. Accountants might use AI for repetitive, background work, such as sorting expense types (an AI may do this with more granularity than human accountants). However, the usefulness of AI still relies on the expertise of the humans who are prompting it and interpreting its output. In the study, senior accountants saw the greatest benefits from AI because they were better able to apply oversight, while junior accountants more often took AI outputs at face value.

### What is AI?

[Artificial intelligence](#) mimics human thought. AI technologies range from small machine learning models that can run on a laptop to large language [models](#) that run on many computers in data centers. Large models [learn](#) patterns from datasets and [predict](#) the next word or pixel based on those patterns. [Generative AI](#) uses these models to generate content that ‘looks right’ based on remixing datasets. While these capabilities can be productively used, for example, to create [concept maps](#) or generate sounds—the ability to generate media and text (i.e. deepfakes) results in harmful media as a form of [digital abuse](#) and [political manipulation](#). [Artists](#) and [musicians](#) have voiced [copyright](#) and creative concerns around AI-generated content.

### Standards of Ethical AI

- Permissibility (Is the usage of AI permitted in class? If so, what are the allowed forms of use approved by the instructor/professor?)
- Integrity (Am I acknowledging AI as a source in my work/the extent to which AI has contributed to my work?)
- [Learning and understanding](#) (Have I learned and understood material myself before using the AI tool? How reliant am I on the AI to complete my work?)
- Accountability (Have I verified the information from the AI tool I used? What biases might the generated prompt contain?)
- Respect for privacy (Is my use of AI respecting my own and others’ privacy?)
- Sustainability (Am I considering the environmental implications of AI use?)

### Harm Reduction Strategies

There are several steps you can take to minimize some of the harmful impacts of using AI tools.

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- [Cite](#) your use of AI and the measures you took to reduce errors.
  - Ex. “This paper uses the AI model [name of tool] for [purpose(s)]. Measures to reduce AI biases, inaccuracies, and other harms included [list the measures used, including time spent, number and design of prompts, etc.]”
- Review AI output for errors, biases, missing information, and false citations.
  - Ex. Verify generated citations are real. Read the generated sources to validate they are appropriate for your research and do not contain biased information. Compare outputs against traditional keyword searches to identify missing information.
- Script prompts to reduce biases in outputs (e.g. include historically-marginalized voices).
  - Ex. “Write two sentences about the greatest discoveries in the social sciences and humanities made by women.”
- Pre-determine the number of prompts you plan to use and scope your research.
  - Ex. Establish the boundaries of your research topic (i.e., cholera in London in the early 19th century) and use [short and clear](#) prompts. Plan an approximate number of prompts needed for your research (i.e., 5-10 prompts to guide literature review).
- Budget the amount of time you use and are exposed to AI content, and take breaks.
  - Ex. Recent research has identified [Generative AI Addiction Syndrome](#) as a concern, so limiting sessions is a potential way to mitigate risks of addiction.

### Cautions:

- Search the tool’s documentation to turn off AI features and opt out of data collection, where possible.
  - Ex. To turn off Copilot in Outlook go to Settings > Copilot and click the toggle to turn Copilot on or off.
- Avoid using AI-generated media to minimize risk of copyright infringement or plagiarism.
  - Ex. Look for visual indicators that media may be AI-generated (i.e., deliberately poor quality, or overly clean with high contrast), or disclosures that content is artificially generated.

## Ethical Issues With AI Use

Appropriate AI use is grounded in understanding the ethical issues surrounding its development and application. Determining your ethical priorities regarding AI can help you make decisions that support your values. We encourage you to use this list as a starting point and conduct your own additional research related to your specific AI use case. Despite the fact that Big Tech

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companies strategically use funding, political power, and other resources to reshape academic knowledge on AI and hide information on AI models, there is growing, critical research on AI.

### Global Impacts

- **Climate:** [Training and using AI](#) require data centers to process large amounts of data. The water and energy used in processing can [worsen water scarcity](#), [increase electricity prices](#), and emit carbon dioxide. For example, It took [50 gigawatt-hours](#) of electricity to train [GPT-4](#), leading to emissions [similar](#) to driving a gasoline car for 50,107,093 miles.
- **Labor:** People from middle- and low-income countries often [labor in poor working conditions with low wages to annotate data](#) for clients in high-income countries. Users of AI tools provide free labor and data to companies, which use their data [to train AI models](#), [target advertising](#), and [sell for profit](#). Companies [use academic researchers and universities as marketers](#), to validate and bolster trust in AI.
- **Violence:** Users of AI tools should be aware that their information can be used for harm. Governments use [AI tools from Google, Amazon, and Microsoft to perpetuate genocide in Palestine](#), a [limited](#) version of [Anthropic's Claude AI to harm school children in Iran](#), and continue violence against historically-marginalized communities.
- **Privacy:** Although AI tools are promoted as having educational features (e.g. transcription and note-taking) they can [inappropriately record confidential information](#) which companies collect for long-term use.
- **Intellectual property and fair use:** Several AI companies are [facing lawsuits](#) from [authors](#) whose content was used without their consent. Some publication venues have made it a [policy](#) that AI cannot be an author and require authors to disclose use of AI.

### Equity and Personal Impacts

- **Accessibility:** AI tools may claim to improve accessibility; although [some of these claims may be deceptive](#). Over reliance on AI tools can have negative impacts on users' [mental](#) and [physical](#) health, and may harm users' abilities to access digital content.
- **Equity:** Data reflects the biases of the society that produced it. These [biases can be amplified when data is in an AI model](#), because they are presented as the "right" answers. In addition, [AI training data is often supplemented by labels](#) added by people, which may contain their [prejudices](#), worsening biases in datasets and model results.
- **Psychological impacts:** Our minds face risks when interacting with AI as a new digital stressor. AI can induce [episodes of psychosis](#), [reduce individual attention span](#), [increase mental exhaustion](#), and [decrease decision-making abilities](#).

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### Learning Impacts

- **Plagiarism:** AI-generated content may [constitute plagiarism](#) and [copyright infringement](#) in some contexts, since it is remixing data that was [scraped from the Internet often without permission from the original creators](#). Review the university's [Policy on the Use of AI Systems](#) for information on approved AI use at Northeastern.
- **Critical thinking:** AI use can [shift your learning experience from creative knowledge production to reviewing AI tasks](#). [Overuse of AI can lead to cognitive fatigue](#), and negatively impact your problem solving abilities. Misuse of AI can also [hinder motivation to self-reflect](#) and evaluate AI outputs, raising issues of academic integrity.
- **Creativity:** The use of AI to augment or replace human creativity has been shown [to reduce users' creative abilities when no longer using AI](#), and [damages users' self-confidence and ability to create ideas beyond AI-generated suggestions](#).
- **Socialization:** Replacing human interaction with AI [can reduce your opportunities for mentoring, peer learning, and group collaboration](#). This can increase isolation and impair your learning experience, since knowledge production often relies on collaboration.
- **AI detection:** Professors are increasingly using AI detection programs, which can falsely flag students' writing as AI-generated. This can be a serious issue for students [who write in multiple languages, are neurodivergent, code-switch](#), etc.

### Resources

- [Philosophy: Technology & AI Ethics](#), Northeastern University Library
- [Northeastern Policy on the Use of Artificial Intelligence Systems](#)
- [Standards for the Use of Artificial Intelligence in Research](#)
- [Northeastern AI Governance and ethics](#)
- [AI Student Guides](#), Center for Advancing Teaching and Learning Through Research
- [Slides: AI for Literature Review](#), Digital Integration Teaching Initiative
- [OECD AI Principles](#), Organisation for Economic Co-operation and Development
- [Data Feminism for AI](#), Lauren Klein and Catherine D'Ignazio
- [Resisting Generative AI Webinar](#) and [Resource Toolkit](#)
- [Social and Ethical Norms in Annotation Task Design](#)
- [Algorithmic Justice League Education](#)
- [BABL AI Online Courses](#)
- [Responsible AI Courses](#), All Tech Is Human